BIRATIONAL MORPHISMS OF DECENT SPACES ARE ISOMORPHIC OVER PAIRED DENSE OPENS

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Editorial note. Counted span is the author's own proof environment, source lines 5281-5285, immediately following the statement at 5268-5279.

Editorial note. The parenthetical '(i.e., integral)' at source line 5273 is retained verbatim; it flags the author's identification of reduced with integral under the decent finite-components hypotheses.

Editorial note. One cross-chapter pointer (Morphisms) and one same-chapter line-indexed stub are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Reduction to schemes via finitely many irreducible components. Lemma lemma-finitely-many-irreducible-components of this chapter, cited at proof lines 5281-5282. May assume X and Y are schemes (maximal representable opens).
- Scheme-side paired dense open isomorphism. Morphisms, Lemma morphisms-lemma-birational-birational, cited at proof lines 5283-5285. The imported terminal statement for schemes.

Source reference. Original lemma 0BB9: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4911–4911. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAC: focused-seeds/geometry/S9-decent-birational-birational/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a birational morphism of algebraic spaces over S which are decent and have finitely many irreducible components. Assume one of the following conditions is satisfied*

(1) *f is locally of finite type and Y reduced (i.e., integral),*
(2) *f is locally of finite presentation.*

Then there exist dense opens $U \subset X$ and $V \subset Y$ such that $f(U) \subset V$ and $f|_U : U \to V$ is an isomorphism.

Proof. By Lemma 0BB9 we may assume that X and Y are schemes. In this case the result is Morphisms, Lemma 0BAC. □

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